INDUSTRIAL WASTEWATER TREATMENT PLANT (IWTP) SUPERVISORY CONTROL AND DATA ACQUISITION (SCADA) SYSTEM

GENERAL

1.01 Project Description

The work to be accomplished consists of furnishing all labor, materials and equipment necessary for a complete automatic monitoring and control system to function as specified herein. The system manufacturer shall furnish a completely integrated Supervisory Control and Data Acquisition (SCADA) system to monitor and control industrial waste treatment processes located in the IWTP compound located at Hill AFB Utah. It shall be the contractor and/or system manufacturer's responsibility to supply a system that is functional and operationally compliant with these specifications. The complete system shall be designed, fabricated, programmed, tested, started up, and warranted by a single supplier (hereafter identified as CONTRACTOR/manufacturer) to ensure a single source of responsibility and support. The contractor may, at their option, employ an electrical installation subcontractor if needed.

1.02 Scope

The SCADA system will be implemented in the following locations/facilities:

IWTP compound located at Hill AFB Utah

CONTRACTOR shall supply a complete and working SCADA system to include but not be limited to the following:

1) All equipment required to accomplish the sequences and I/O schedule.
2) All labor for installation and start-up of the system.
3) Submittal and shop drawings prior to installation.
4) All ancillary equipment, hardware, software, and appurtenances needed for proper installation and operation of the SCADA equipment.
5) Provide spare parts and maintenance tools as described below.
6) Operations and maintenance manuals as detailed below.
7) All start-up labor and services.
8) All operator and maintenance training.
9) Remote Interactive support tool (software and/or hardware) shall be provided. The current pre-approved remote Interactive support tool is Infoscan.

OWNER will supply electricity, access and easements as needed for all sites, and desks, furniture and chairs for any computer equipment, unless otherwise stipulated in the specifications. Owner will also supply and maintain any re-used equipment and devices including data transmission media and media converters as applicable.

1.03 Quality Assurance

The specifications and drawings detail the salient characteristics and required functions of the SCADA system and represent the minimum acceptable standard of quality for equipment, materials, support and methods of construction required for this project.

1.04 Manufacturer's Qualifications and Basis of Design

The system specified herein shall be the product of a manufacturer who can demonstrate at least 15 years of satisfactory experience in designing, manufacturing, furnishing, installing, and supporting comparable SCADA for water and wastewater treatment, distribution or collection systems.

The manufacturer of this system shall maintain an available inventory of all replaceable SCADA modules to assure the Owner of prompt maintenance service and a single source of responsibility. The manufacturer shall certify availability to the OWNER in writing in their proposal. Field installed microprocessor-based controllers used in the SCADA system shall be guaranteed serviceable for a period of not less than 7 years from warranty start date. This includes repair and/or replacement parts. Manufacturers that cannot certify availability of service will not be acceptable.

The basis of this design and specification is the Info-Scan SCADA system, manufactured by Dorsett Controls (Dorsettcontrols.com)

Substitution of equipment by other manufacturers will be considered for Pre-Bid approval if equipment proposed for substitution is demonstrably equal or superior in quality and efficiency to the standards established in the specifications. Suppliers and contractors wishing to substitute alternate manufacturers' equipment shall submit a compliance checklist for each item and requirement in this specification for the item to be substituted. Proposed substitutions without compliance checklists will be rejected without discussion. The determination of acceptability is solely the responsibility of the ENGINEER and OWNER.

Requests for Post-Bid Approvals of substitute or "or equal" equipment will not be considered.

Other manufacturers' equipment must be submitted for pre-bid approval in accordance with the pre-approval requirements listed in this specification.

1.05 Codes and Standards

The control system and its components shall comply with all applicable requirements of the following:

A) National Electrical Code (NEC)
B) NEMA Compliance
C) IEEE Compliance
D) EIA Compliance
E) UFC 4-010-06
F) DoDI 8500.01 Cybersecurity
G) DoDI 8510.01 Risk Management Framework (RMF)

1.06 Submittals

Complete submittal shall be provided to CEOI or designated party for approval prior to equipment fabrication. The intent of the submittal requirement is to allow the owner to ascertain the suitability of the proposed system to the project requirements and to allow the Subject Matter Expert (SME) to review compliance with the specifications. The submittal data shall include the following as a minimum:

A. Product Data: Provide product data sheets for each instrument and component to be supplied in the system. The data sheets shall show the component name as used on reference drawings, manufacturer's model number or other product designator, input and output characteristics, scale or ranges selected, electrical or mechanical requirements, and materials compatibility.

B. Shop Drawings: Provide drawings for each panel showing the wiring diagrams for control circuits and interconnections of all components. The drawings shall include wiring diagrams for all remote devices connected to the panel.

C. Panel Layout Drawings: A front panel and sub-panel layout shall be included as part of each control panel drawing. Components shall be clearly labeled on the drawing.

D. Installation Drawings: Typical installation drawings applicable to each site in the system shall be included.

E. Operator Interface Software: The submittal shall include a generic but detailed technical description of the Graphical User Interface software (GUI) as proposed for this system including:

Sample text screens and menus
Sample graphics screens
Sample report logs and printed graphs

1.07 Maintenance Information and Requirements

A. Operation and Maintenance Data Manuals: Upon completion of system testing and owner acceptance, submit operation and maintenance manuals and "as built" drawings on all items supplied with the system. The manuals and drawings are to be bound into one or more books as needed. In addition to "as built" engineering submittal data and drawings, the manual shall include operation and troubleshooting guides and maintenance and calibration data for all adjustable and serviceable items.

B. Maintenance Stock: The System Manufacturer shall also provide 10% spare fuses and lamps for each type used in the system. Provide one each spare part to include:

(1) of each type of modem or communications interfaces.
(1) of each type of RTU Power Supply.
(1) of each type of RTU Controller.
(1) of each type of Input / Output Relay.
(1) of each type of Signal Conditioning device

No spare parts will be required for the Central Server Computer, LCD monitor, printer, passive backplanes, or wiring harnesses.

1.08 Project Conditions

All instruments and equipment shall be designed to operate under the environmental conditions where they are to perform their service. The equipment shall be designed to handle minimal transient voltages as normal environmental hazards. A direct lightning flash/strike in the vicinity of the equipment is considered an act of nature and not covered under product warranty, however any failed equipment which does not exhibit visual evidence of a lightning strike and was not installed with surge suppression devices must be covered under warranty. The environmental conditions are as follows:

A. Outdoor: The enclosure equipment will be exposed to direct sunlight, dust, rain, snow, ambient temperatures from -32 to +122 degrees F, relative humidity of 10 to 100 percent, and other natural outdoor conditions.

B. Indoor: The equipment will be capable of operating at ambient temperatures of +32 to +122 degrees F and relative humidity of 20 to 100 percent.

1.09 Delivery, Storage and Handling

All items shall be stored in a dry sheltered place, not exposed to the outside elements, until ready for installation. All items shall be handled with appropriate care to avoid damage during transport and installation.

1.10 Sequencing and Scheduling

A. Coordination: The contractor shall coordinate with electrical and mechanical work including wires/cables, raceways, electrical boxes and fittings, controls supplied by others, and existing controls, to properly interface installation and commissioning of the control system.

B. Sequence: Sequence installation and start-up work with other trades to minimize downtime and to minimize the possibility of damage and soiling during the remainder of the construction period.

PART 2.00 - PRODUCTS

2.01 SCADA Control System Description

A. General: The control system shall use on-site and/or remote RTU controllers, hereafter referred to as “controllers”, to interface to all sensors and mechanical equipment as applicable. The system shall be a "distributed intelligence" type control system that provides local decision-making capability in the local/remote controllers if local stimuli are available to the control application. Additionally, controllers shall have the ability to utilize data from outside its own input/output map in order to automatically control processes and equipment. The system operator shall have the ability to designate the residence of all control applications based on best process control practices. Generally, all functions available at the central server shall be available at the local/remote controllers, provided the local I/O supports the functionality. The system shall also support combinations of server/local/remote (stand-alone) sequences as required to affect a unified control system.

The control system shall be capable of utilizing several means of communication with controllers as well as networked computer equipment. Generally, as a minimum the following methods are required:

High-Speed Ethernet networked communications over copper or fiber

High-Speed Serial data communications over fiber or copper

Private Network over Cellular communication providers

Each controller panel utilizing Ethernet shall be assigned a unique IP address. Coordinate with owner if integrating the existing municipal LAN for valid IP addresses. The contractor shall provide all required devices for the chosen method of communications as best suits the locale and existing conditions.

The applications programming used at all controller locations shall be stored in non-volatile memory that are both server and field programmable using software detailed later in these specifications. The controller devices shall be "self-initializing" and not require operator intervention after power interruptions, transients from lightning storms, or applications programming control logic changes. All controllers in the system shall include "power-fail restart" circuitry to ensure automatic restarts of the system. The central server shall retain a complete copy of all controller configuration data and operational parameters such that an automatic re-configuration of all controllers is possible at any time, as initiated by the operator. Additionally, the central server software shall be capable of downloading any setpoint or modified I/O point data including modified applications programming individually to any networked controller without disruption of the other controllers or processes. This capability shall not be impacted by the form of communications chosen for the system.

The central server (CS) shall address each remote site as needed, to send command data and configuration changes. The CS will receive status data from each remote site in a report by exception method or as values change. These values are in the form of status, alarm, flow, pressure, level, and other data as required from each site. The server database shall automatically record all information received from the field and permit indication, display, alarm, and operator interaction via the Operator Workstation. All system information shall be readily available at the workstation provided the proper security levels are met by the operator.

B. <u>Controller Software Features:</u> The controller control algorithms shall have the ability to integrate both hardware and software operator inputs, along with hardware inputs at the remote sites into a cohesive automatic operating control system. At a minimum the logic shall provide the following features:

Monitor input conditions at local and remote controllers to determine the validity and/or usability of process input signals. Additionally, determine the correctness of generating discrete commands (i.e. monitoring high discharge and low suction cut-off controls at pump stations) before starting or running equipment, adhere to minimum on/off times, and control priority resolution.

2.02 <u>RTU Controller "Distributed" Control Software Features</u>

All controllers shall be capable of executing local control in coordination with server supervisory operations. Setpoints for "distributed" control at the controllers shall be operator selectable from the server, operator workstation.

The controller shall be able to implement all pre-determined required modes of control in the event of server or remote communications failure. In the unlikely event that the server fails, a back-up server will act as the server until the server is repaired.

The controller shall be capable of receiving new or modified programs from the server without requiring field modifications to the RTU hardware or software. Systems that require “pre-loading” or local loading of controller I/O database or local control logic programming are unacceptable.

The controllers shall be capable of locally storing change-of-value data for any desired points. Whenever a point’s value changes or its quality status changes, the controller will store the new value and status with the current time and date. For a digital or binary type point, a change-of-value is detected for any value change. For an analog type point, a change-of-value is detected when the value changes by more than the definable change-of-value (COV) delta. Each controller shall have the capacity to store values for up to 60,000 changes. These stored values shall be maintained in a first-in / first-out buffer.

PART 3.00 - SERVER EQUIPMENT AND SOFTWARE:

The system shall be capable of utilizing on-premise server hardware for the central server (CS) role and functionality. No loss of functionality shall exist for either platform with respect to the herein specifications.

Windows Server 2025 will be the server operating system and Windows 11 for OWS.

The "SCADA System" shall be composed of a minimum of 2ea separate servers communicating over a high-speed Ethernet or other type data link. The primary server computer shall initiate and/or process all communications with remote units and operator workstations. The server shall also be capable of acting as an operator interface to the system and provide display, alarm, reporting and logging of all data. The 2ea computer (called the Operator Workstation or OWS) will serve as the primary interface used to control and monitor the process equipment.

3.01 System Hardware and Software Descriptions

A. Hardware -General: The server and OWS computer hardware shall be the standard data processing equipment of a single manufacturer and not be modified in any way that would preclude the continuation or purchase of the manufacturer’s standard support contracts. This shall include the processor, display, keyboard and other “input” devices.

B. System Processor Unit: The server and workstation computers (if provided) shall be an Intel/Windows based processor. It shall include as a minimum the following features:

1) Intel Xeon Quad Core
2) 32 GB RAM
3) Hardware RAID controller "Raid 1 Configuration"

4) Dual 960GB SSD SATA Hot plug
5) Hot-Plug Redundant Power Supply
6) Dual Port 1Gb On-Board NIC
7) On Board Adapter or Graphics Card
8) USB Sound Bar or USB External Speakers
9) Windows Server 2022 Standard
10) 2 Button Optical Mouse
11) QWERTY Keyboard with numeric keypad and cursor keys
12) DVD+/-RW SATA Internal
13) ProSupport and 4Hr Mission Critical 36 Months

The system unit shall be housed in an enclosure appropriate for the environment in which it is to be installed. Generally, this could be rack mount if being added to an existing server or IT rack, or desktop tower if a completely stand-alone local SCADA-only server. The server will be selected for the SCADA application software, database and designed environment, plus 50% expansion capacity. The server shall store all databases, graphics, backgrounds, command inputs, and set points. The main disk drives shall be mirrored (specify RAID level) for fault tolerance, data backup and retention purposes. In the event of primary disk failure, the mirrored drive shall be available for operation. The USB Flash Drive shall be used for archive data storage and back-up protection of the operating program. It is recommended that one flash drive be taken off site with the latest downloaded data. An optional cloud backup option should be available that will retain one monthly backup and a rolling daily backup storing up to seven days.

The separately mounted keyboard shall have a standard typewriter format with tactile feedback, twelve special function keys, and a separate numeric keypad for entering set point data and cursor control. The 256-character symbol set shall include 96 ASCII characters and the IBM (International Business Machine) graphic symbols. The system shall include a 2-button optical compatible mouse.

C. <u>Accessories:</u> The computing equipment (including LCD display, Keyboard, and System Unit) shall be located on top of a desk or other appropriately sized platform supplied by the Owner. All interconnecting cabling shall be plug-in and supplied by the contractor.

D. Battery Back-up Operation: The server computing equipment shall include a minimum of 20 minutes of backup power. The back-up units shall be a Standby Uninterruptible Power Supply (UPS) system that provides power line filtering and transient protection. The unit shall automatically transfer power when the power line fails without interrupting or restarting the system and automatically recharge the battery within 10 hours after the power returns to normal. The UPS shall be located at the location of the computers and shall power the System Unit, all communications equipment, modems, network equipment, LCD display and printer in order to maintain uninterrupted system operation. The UPS shall be sized as required to supply the selected server and ancillary devices for 20 minutes backup minimum. Provide calculations in the submittal documents prior to purchasing equipment for owner/engineer review.

E. Notification / delivery Systems: The system shall allow end users to be notified using the internet by the following methods: email, SMS (text), or Voice call. Most notifications/delivery systems rely on 3rd party providers to relay all emails, SMS (text), or Voice calls. The notification/delivery system is a two-part system; the first part is responsible for delivering alarm status to end users in sequences defined within the system. All delivery methods are available for this part. Also, the end users may have the ability to acknowledge alarm status via emails and voice calls depending on system setup and customer needs. The second part is responsible for delivering reports at scheduled times to end users via email.

F. System Back-up & Installation: The contractor shall provide a back-up copy of the installed software on a CD-ROM or DVD disk or USB Flash Drive. Back-up copies of any setup or graphic files shall be on a CD-ROM or DVD disk or USB Flash Drive. The copies shall be kept by the Owner for emergency reloading in the event of a catastrophic failure. A cloud backup option shall be available that will retain one monthly backup and a rolling daily backup storing up to seven days. Restore option will be available through the GUI interface for any saved backup.

H. System Restart: The software and hardware shall automatically restart in the event of a power failure. All data necessary for the operation of the systems shall be reloaded from memory.

I. System Capacity: At a minimum, the application software shall be fully licensed and capable of accommodating 100,000 I/O points as follows:

1) Binary inputs and outputs
2) Analog inputs and outputs
3) Accumulated inputs
4) Multi-state inputs and outputs
5) Calculated control points

The system shall also be capable of supporting the following:

1) Operator workstations (up to a maximum of 12 local or remote)
2) Main Menu page (with direct access to all screens and other program features)
3) System Summary page listing key data points for all controllers on system
4) Display pages showing all data for each local/remote panel in the system
5) Remote connection via secure internet connection for service and support
6) Multiple protocols including Modbus RTU & TCP, BACnet MS/TP & IP, and N2
7) Permissions - Users are only given access to perform functions they are authorized to perform.
8) Permissions – Users may only access functions at the authorized time or day or day they are authorized
9) Limit unsuccessful logon attempts.
10) Provide user privacy and security notices consistent at login
11) Use session lock with pattern-hiding displays to prevent access/viewing of data after period of inactivity.
12) Terminate (automatically) a user session after a defined condition.
13) Create, protect, and retain information system audit records
14) Limit information system access to authorized users, processes acting on behalf of authorized users, or devices
15) Prevent reuse of username identifiers for a defined period.
16) Disable users after a defined period of inactivity.
17) Enforce a minimum password complexity and change of characters when new passwords are created.
18) Prohibit password reuse for a specified number of generations.
19) Allow temporary password use for system logons with an immediate change to a permanent password.
20) Store and transmit only encrypted representation of passwords.
21) Require multifactor authentication to establish nonlocal maintenance sessions
22) Implement cryptographic mechanisms to prevent unauthorized disclosure of data during transmission
23) Protect the authenticity of communications sessions with SSL certificate
24) Ensure that the actions of individual system users can be uniquely traced to those users so they can be held accountable for their actions.
25) System to provide audit reduction and report generation to support on-demand analysis and reporting.
26) System must require multifactor authentication for access
27) System must obscure feedback of authentication information.as to not indicate that user accounts exist.
28) System must provide separation of the duties of individuals to reduce the risk of malevolent activity without collusion.

Beyond the basic application software required for SCADA operations, the software package shall accommodate the following:

J. Analog Data:

1) Display value directly in engineering units
2) Accept operator adjustable High & Low alarm limits and generate alarms
3) Store data for trending displays

K. Flow Rate Data:

1) Display value directly in engineering units
2) Accept operator adjustable High & Low-rate alarm limits and generate alarms
3) Calculate flow total and display in engineering units
4) Accept operator adjustable High/Low 24-hour total limits and alarms
5) Store data for trending displays

L. Pump Control Operations:

1) Display RTU control and status functions
2) Display Pump alarms and status for each pump
3) Input/Display control database

M. Status Point Operations:

1) Display RTU status functions
2) Input/Display control database

N. Alarm Point Operations: The SCADA system shall produce alarm messages for points that are outside of their normal operating parameters, points returning from an abnormal state, and network events (such as device configurations). Each alarm message shall consist of an alarm class indicator, the date and time at which the alarm occurred, the point name that produced the alarm, and the alarm message text. Several methods for reviewing alarms shall be provided. They include:

1) The Alarm window presents all alarms to the operator for acknowledgement. The purpose of this window is to draw attention to alarms that may need operator attention.

2) The Recent Alarms window presents the most recent alarms in descending order newest to oldest by default. The purpose of this window is to allow the operator to monitor system events and alarms in their chronological order.

3) The Active Alarms window shows alarms that have not yet returned to normal. The purpose of this window is to show those alarms in the system that has not yet been corrected. When the point showing alarm returns to normal, the alarm is removed from the Active Alarm window.

4) The Alarm History of the system allows the user to review all past alarms. In each of the above views time, date, and point name can be used to filter the alarms as well as class and category when needed. There is no requirement for alarms to be archived from the database to disk files or other archiving methods. However, if archiving is enabled, all archived alarms shall also be available for review within the Alarm History views from above.

O. <u>Event Point Operations: The</u> RTU controller databases shall be configured with change of value (COV) report delta values per the owner's requirements. The COV delta will be used to determine the amount of change in value a point must experience before it is required to report a new value to the server. Generally, each point will have COV deltas appropriate for the expected variances in normal operation. This is designed to minimize network traffic and optimize system resources. The COV delta shall not be used in control loops and use actual values instead. For any COV event the following functions shall be performed:

1) Display CS, and RTU alarm functions
2) Display CS, and RTU event functions
3) Enter new event in data log archive
4) Log event clearing and send alarm clear to database
5) Input/Display control database

P. <u>Database Editor:</u> The Database Editor shall be an on-line software program, integral to the application software for the generation and modification of the database for the system. The database shall include all non-windows data that comprises the SCADA system. The contractor shall generate and install the initial database for the SCADA System, including the graphics displays from the I/O tables, sequences, reports and other owner criteria. It shall be possible to perform the editing function while the system continues to poll remotes for data and performs its normal SCADA control functions. The editor shall operate in English and shall not require an understanding of programming as a prerequisite for its use or modification. The editor shall allow for the creation and of text and graphic displays pages. At a minimum, the following features shall be available to create or modify the database:

1) Status Point Definition

a) Add, delete or modify individual points including the following specific characteristics:
b) Status points logic
c) Point and Station name
d) Alarm conditions
e) Normal state
f) Designation of multiple status points associated with a single control point to allow for verification of proper operation
g) Cloning of database items for faster database creation

2) Analog Input points

a) Point and Station name
b) Scaling and calibration factors
c) High/Low alarm limits
d) Normal Condition
e) Cloning of database items for faster database creation

3) Text Display Pages

a) Alphanumeric, line and special functional characters for Text Displays
b) Status/Control points (On/off/alarm display)
c) Foreground/Background color selection

4) Color Graphic Displays

a) The graphics development package shall include predefined symbols, allowing the operator to generate custom symbols, and provide for importing third party windows format graphics. The status of the equipment shall be indicated by the color of the symbol shown on the display. The following colors shall be used consistently in all displays to indicate the status of equipment:

High Alarm - Red
Low Alarm - Blue
Failed Alarm - Yellow
Out of Service/Manual - Green

5) The Contractor shall generate a custom color graphic display for each grouping of equipment in the system. For example, a graphic display will be required for each building's processes in the system as well as the overall plant. Animation shall be included for moving equipment and process media.

6) The system shall provide for password protection to any desired area of the system. Each operator shall be assigned a unique password to control his access. The system shall utilize Two-Factor Authentication via authenticator application. The system shall restrict password reuse and disable accounts due to inactivity. Accounts will be temporarily deactivated after 5 invalid login attempts. The system shall day and time restrict user access based on user account or role. At a minimum, the system shall be capable of limiting access to certain user-designated areas:

a) Engineering Level – access to system setup parameters
b) Supervisory Level – access to start/stop and high/low alarm settings
c) Operator Level – access to equipment controls

Q. <u>Real-time Graphic Displays:</u> The system shall include real-time custom graphic displays. The graphic displays shall be multi-color dynamic graphics with accurate representation of all piping, pumps and ancillary equipment. All data displays of the measurement point shall be color designated based on which limit has been violated (e.g., black-normal, red-emergency, etc.). If animation of moving equipment is required by these specifications, it shall be included on these displays. Animations shall be capable of a minimum of four distinct state animations, i.e., On-normal, off-normal, on-alarm, off-alarm. Each state shall be represented by a different animation or color frame.

R. Data Archiving: The program shall track all the activity in the system. The program shall retain and store the disk system, the last value for each input/output as well as any historical profile data desired by the owner. The archived history data shall be stored online and offline as required to retain data permanently. The CS shall automatically backup data onto the mirrored drive. A removable storage device should be used not less often than weekly, and more often as determined by the system manager to store backup data offline and securely off-site for safekeeping. The data shall be directly usable by the system in its compressed form from either the hard disk or removable storage media. The historical data storage capacity shall be limited only by the amount of storage space on the system drives.

S. Automatic Logging: The system shall automatically log all alarm and event activities (i.e. occurrence, acknowledgement, and clearing) and operator changes to the system. In addition to the event and alarm data, the system shall log all operator actions into a permanent indexed and searchable operator's activity log. This log will include all operator interaction with the system to include modifications to points, command to points, changes to reports and any other database modification including displays. The log shall be searchable by operator, point name, date and time. The report shall be exportable to a comma-delimited ASCII file format for import into Excel or other database program.

T. Historical Data Trending Module: All data is to be accumulated on the hard disk as either online or archived history data. All historical data shall be backed up when backups are performed by the system or operator. Archived data shall be retained for as long as storage capacity exists on the drives.

U. Trend Module: The Trend module shall allow for the short-term monitoring of data points at intervals ranging from 1 minute to 24 hours. The system shall have the capacity to run multiple independent Trend Plots on a single operator workstation at the same time. The data points history resolution can be collected at up to 1 second and refreshed on the trend plot at least every 20 seconds.

W. Auto Trend: Auto Trend data shall be collected and stored by the RTU controllers and be displayed via the main database selector tool for any point that supports automatic trending. The samples shall be displayed along with their time stamps and status information (Out of Service, Overridden, Fault, and Alarm).

X. Historical Report Generator Module: History Logs shall allow long-term monitoring of up to sixteen data points at sample rates from 5 minutes to 24 hours. Data shall be collected continuously by the server and stored indefinitely. The number of history logs stored on the system shall be limited only by available disk capacity. Report tools include a tabular report for a user defined date / time range as well as a charting tool to plot the data in graphical format. The system shall include a Historical Report Generator module that allows for operator selected data to be pulled from the online or archived database and converted to a comma separated value (CSV) file format. The format shall allow for data import into other software packages for data review by the Owner or his engineer. The data search shall allow for all archived and accumulated data to be searched.

Y. <u>Graphical Sequence Programming Tool:</u> A Graphical Programming Language (GPL) tool shall provide users with a graphical interface which has the capability to create, edit and display control sequences. Users shall be able to rename, add, clone, or delete a GPL sequence. GPL Sequences shall only be created by operators with sufficient access levels to do so. The operators shall be able to graphically build their sequences by connecting lines and function blocks graphically, similar to a CAD program drawing. Once a sequence has been added, it may be compiled, activated in the RTU and saved. A user can then display the sequence with all of its associated values running in dynamic real-time. The following graphic programming blocks shall be provided as a minimum:

1) Alarm Status Block
2) Average Block
3) Summation Block
4) Digital (Binary) Mode Control Block
5) Comparator Block
6) Delay Block
7) One Shot Block
8) Digital Logic Block
9) Find Largest or Smallest Block
10) Logic Block
11) Math Block (basic math operator plus square root and totalizer features)
12) Analog MUX Block
13) Digital MUX Block
14) Proportional / Integral / Derivative (P only, PI, PID) Block
15) Setpoint Adjustment (SPA) Block
16) Analog Mode Control Block
17) Ramp startup Block
18) Totalizer Block
19) Building Automation Blocks (Dewpoint, Enthalpy, Economizer, wet bulb)
20) Digital Logic Blocks (And, OR, XOR functions)

Z. <u>Remote/Mobile Access:</u> The SCADA system shall be accessible from any Internet connection when required by the owner. All user to server sessions shall be encrypted and use industry standard security measures between any user device and the server.

From a phone or tablet web browser with the required security credentials provided by the user shall support the monitoring of the RTUs and alarms including acknowledge, as well as issuing basic commands to equipment.

From a workstation the user shall be able to install the client software from a download page served by the server. At a minimum all current Microsoft supported Windows operating system versions shall be supported by the client software. The client software shall allow full access and full features for the entire system.

The internet service will be provided by the owner. The Owner or Contractor will provide any required routers and equipment required to provide the remote mobile access capability unless otherwise stated.

PART 4.00 – LOCAL AND REMOTE CONTROL UNIT I/O EQUIPMENT

A. <u>General:</u> All on-site and remote controllers shall be equipped with "smart" programmable controller units at all locations. The core software program used at all locations shall be identical for each controller type and stored in non-volatile FLASH type ROM memories that can be upgraded from the server or OWS or remotely by authorized personnel with proper security level. The core controller software shall provide the basic operational logic including communication with other controllers in the system and responding to control commands from the server.

Control and configuration data shall normally be stored in battery-backed Random Access Memory (RAM) or flash type memory for use by the controller's processor. All controllers shall be completely programmed via the server download and without the use of external adapters or supplemental programs or data. The RTUs shall include "watch-dog" circuitry and be "self-initializing" without operator intervention. In the event that the program or configuration data is corrupted, the server shall reload the program and configuration data as long as communications are still established.

In addition to the GPL control functions described in section 3.Y (GPL), the controllers shall also support a process script language for site-specific customizations including special input and output manipulations, local sequential control as follows:

1. Arrays
2. Comments (Full line, In line)
3. Variable Types (Boolean, Integer, Real)
4. Arithmetic Operators (Addition, Subtraction, Division, Multiplication, Modulo Division, Exponentiation)
5. Relational Operators (Less than, Less than or equal, Greater than, Greater than or equal, Equal, Not equal)
6. Logical operators (AND, OR, NOT)
7. Bitwise Operators (Shift, AND, OR, XOR, NOT)
8. Arithmetic Functions (Absolute value, Maximum value, Minimum value, Square root)
9. Logarithmic functions (Natural log, Log base 10, Exponentiation Base e)
10. Procedures
11. Conditional (If, Then, Else)
12. Loops (For, Do, While)
13. Time Functions (Time, Date, Day of Week)

B. <u>RTU Construction:</u> The RTU shall be deemed to include all power supplies, controllers, communications modules, and basic inputs and outputs.

The controllers shall have a minimum of three communication ports: Two serial and one ethernet. The serial shall support minimum baud rates of 9600 up to 76800 baud and have a plug-in connector that provides a 2-wire RS485 interface. All the ports shall also have LEDs to allow monitoring of the communications activity. The ethernet shall support fast Ethernet for communications to the server and other RTUs located on the high-speed network. The Ethernet port shall conform to all EIA standards for fast Ethernet. Surge protection shall be provided for all ports leaving the originating building. These specifications may apply to RTUs when local controllers are utilized in a physically remote location such as an RTU but are considered part of the local network.

All local and remote panels shall provide for sufficient installed and configured spare inputs and outputs (I/O) to meet the site requirements as detailed and provide for 10% spares of each type. Owner use of the spare capacity shall not require the addition of hardware or any monetary charges to the customer for licensing or additional software modules.

All controllers used in panels shall be easily replaced in the event of failure without special programming, removal of wiring, or other components in the panels. Removable terminal blocks or electronic modules are required.

The controller shall employ multiple 32-bit processors running at speeds up to 180MHz. Hardware DMA and deep communication buffers allow card

communications with minimal processor intervention, freeing the processor to concentrate on the end user application. Including a minimum of 4MB of battery backed RAM paired with a 4.5MB of non-volatile flash to allow program growth and data storage. At minimum a 2-line LCD & keypad interface allowing interrogation of the I/O status, system health, and hardware setup information.
The controller shall be capable of terminating input and output based on specifications below.

Digital Outputs
<u>120VAC 16 Channels</u>
- Output Voltage Rating 90 – 130 VAC
- Max Amps per Bank 680 mA
- Max Amps per Channel 125 mA
- Min Channel Amps 5 mA

<u>24VAC 8 Channels</u>
- Voltage Output 4 24 VAC
- Per Channel Current Output 300 mA (max)
- Total Output (all channels) 2 A (max)

<u>12VDC 8 Channels</u>
- Isolated Source-Sink Output (1W) 30 mA(max)
- Sink Only (0.8W) 6 100 mA(max)

<u>Relay 5-8 Channels</u>
- Form A and C
- Max Operating Voltage Range 240 VAC / 125 VDC
- General Use (Resistive Load) 2A
- Contact Rating Code "See manufactured chart for voltage range)

Analog Outputs
<u>0-20ma 4-8 Channels</u>
- Accuracy 1 ±120 uA
- Analog Resolution 10 Bits
- Load Impedance Max 600 Ω

<u>0-10VDC 8 Channels</u>
- Resolution 10 Bits
- Voltage Output 0 – 10 V
- Resistive Load 600 Ω (max)

Analog Inputs
<u>4-20ma 2-16 Channels</u>
- Loop and Self powered options
- Accuracy ±20 uA
- Analog Resolution 16 Bits
- Input Voltage Burden Max 4 VDC
- Loop Supply Voltage 24 (±15%) VDC

<u>0-10VDC 16 Channels</u>
- Analog Resolution 16 Bits

- Voltage Input 0 – 10 V

0-2k Resistive 16 Channels

- Analog Resolution 16 Bits
- Resistive Input 0 – 2k Ω

Digital Inputs

120VAC In 5-16 Channels

- Max Input Voltage Rating 130 VAC
- Min 'ON' Voltage 90 VAC
- Max 'OFF' Voltage 30

Relay in "Dry contact" 4-16 Channels

- Typical Open Contact Voltage 10-24 VDC
- Max Open Contact Voltage 30 VDC
- Max Closed Contact Current 1 mA
- Maximum Input Frequency 1 200 (50% duty cycle) Hz

Power Supply

- Each RTU assembly shall include an integral power supply. Power supplies shall be designed for 120 VAC input power and suitable for use with UPS operations.

C. Enclosures: The unit enclosures for indoor mounting shall meet all the requirements for NEMA Type 12 enclosures. The enclosures body shall be made of a minimum 14-gauge steel with continuously welded seems and be furnished with external mounting feet. The enclosure door shall be made of minimum 16-gauge steel with a 14 gauge steel hinge. Enclosures larger than 16x14 shall have a rolled lip on 3 sides of the door for added strength. The door opening shall have a rolled edge on 4 sides to protect the door gasket. The door gasket shall be heavy neoprene and attached to the door with oil resistant adhesive. Sub-panels shall be 14-gauge steel for 16x14 enclosures and 12 gauge for larger enclosures. The enclosure finish shall be gray polyester powder coating inside and out over phosphatized surfaces. The subpanels shall be finished in white. Nema 12 enclosures shall be Hoffman "CH" or "CONCEPT" wall mount enclosures or equal. Alternately, fiberglass enclosures may be furnished if they meet the specific NEMA requirements and are a standard manufacturer's product.

Remote site installations requiring equipment to be mounted outside shall have a lockable NEMA 4X 306 Stainless Steel enclosures. The enclosure shall be required to control vandalism, provide complete weather protection, reduce the heating effects of the sun, and prolong the life of the equipment. The enclosure shall be constructed of 14-gauge stainless steel, with a drip shield top and seam-free sides front and back, and a stainless steel hinge pin. The NEMA 4-X enclosure shall be Hoffman or equal. Alternately, fiberglass enclosures may be furnished if they meet the specific NEMA requirements and are a standard manufacturer's product.

The remote unit enclosures that are mounted on exterior racks & corrosive areas (such as concrete meter vaults) shall be NEMA Type 4X rated enclosures. The enclosures shall be made of stainless steel or molded fiberglass polyester and be furnished with external mounting feet. The door shall have a seamless foam-in-place gasket and corrosion-resistant hinge pin and bails. Sub-panels shall be 14-gauge steel for up to 16”x14” enclosures and 12 gauge for larger enclosures. The subpanels shall be finished in white. Nema 4X enclosures shall be Hoffman, Saginaw, Vynckier, or equal.

D. <u>Control Functions:</u> In general, the local/RTUs shall be programmed to provide generic and specialized control functions as detailed earlier and to work in concert with the server to completely accomplish the specified sequence of operations and operational criteria of the owner... The contractor shall be responsible for meeting with the owner and the engineer to develop the automatic control strategy required for the system to perform as specified. All programmatic implementations of sequences shall be provided to the owner for approval in the submittal package described earlier in this specification.

<u>PART 5 – DATA COMMUNICATION</u>

<u>5.01 The SCADA system will utilize a high-speed communications network to facilitate the collection and sharing of data within the system. A combination of local and/or remote communications methods may be required.</u>

A. Ethernet over Copper - Communications channels shall be provided as shown. Each server or OWS computer shall be supplied with an internal network interface card for connection to IEEE STD 802.3 BASE-T twisted pair Ethernet LANs. Interface cards shall be supplied with an on-board transceiver for direct connection to the LAN, and with an AUI port for performing diagnostics. Interface cards shall also have an on-board buffer memory to prevent loss of data packets. The LAN shall interconnect and service system local and remote components. The network transmission media shall be Category 6 twisted pair cable as defined by EIA ANSI/TIA/EIA-568-B. All cabling, patch panels, patch cables, and accessories shall be provided as required to implement a complete wiring system for the LANs. Connector type shall be Category 6 rated. Connections to the controllers using Ethernet will meet identical wiring standards as above.

B. Ethernet over Fiber - Communications channels shall be provided as shown. Each server or OWS computer shall be supplied with an internal network interface card for connection to IEEE STD 802.3 BASE-T twisted pair Ethernet LANs. Interface cards shall be supplied with an on-board transceiver for direct connection to the LAN, and with an AUI port for performing diagnostics. Interface cards shall also have an on-board buffer memory to prevent loss of data packets. The Ethernet over Fiber-Optic LAN shall interconnect and service system local and remote components. The network transmission media shall be single or multi-mode fiber optic cable suitable for installation in the required environment. A minimum of six fibers shall be installed to each location and terminated in patch panels with LC, ST, or SC connectors. Provide all patch cables to connect to the provided Ethernet to Fiber-Optic media converters. All fiber optic cable shall be suitable for high-speed (1 Gb) or greater transmission speeds. All cabling, patch panels, patch cables, and accessories shall be provided as required to implement a complete wiring system for the LANs. Locate media converters in areas safe from potential fiber damage. All media converters, hubs, switches and similar devices must be connected to backup power sources. Connections to controllers using Ethernet will meet identical wiring standards as above.

PART 5.00 - EXECUTION

5.01 Equipment Testing

The control system components shall be completely tested prior to shipment. The central server and local/remote RTU panels shall be tested at the factory for a period of at least 3 days.

5.02 System Start-Up

The manufacturer shall supply "Factory" personnel for start-up service as needed to insure satisfactory operation. Subsequent trips to the job site to correct defects shall be made at no charge to the Owner during the warranty period. Start-up shall be a planned and well-executed event minimizing downtime of the plant processes and reducing switchover to the new SCADA system. The switch over plan shall be communicated to the customer at least 2 weeks prior to switch over and be scheduled with the customer.

During the start-up and switchover work, the customer/owner shall be permitted to observe and/or assist in the work to facilitate a better understanding of the system as part of the training effort. The training will not be time limited but will be permitted any time during this process. The environment shall be considered informal.

5.03 Training

The system manufacturer shall supply "factory" personnel to conduct on-site training sessions, totaling a minimum of seven business days of training for covered shifts.

The initial training session shall be conducted during start-up as needed until the Owner and Engineer are satisfied that the operators are comfortable with the operation and maintenance of the system. Training shall be done on site with the owner's personnel. The training will consist of instruction in areas of basic workstation operation, monitoring system, preparing reports, control of system operation, hardware training, controller replacement, hard system resets, and general system maintenance.

Three months after the Owner commences system operation, the system manufacturer shall supply "factory" personnel to conduct follow-up training of the Owner's personnel. The follow-up training shall be conducted on-site and consist of reviewing the operation and maintenance of the system, adding I/O to the system, modifying graphics, and modifying and adding control sequences. The Owner shall be contacted a minimum of two weeks in advance, prior to scheduling the training session to allow proper coordination.

5.04 Substantial Completion

The owner/engineer will grant substantial completion status only after completion of the start-up and initial training phase of the project. The Engineer shall make an inspection of the system to determine the status of completion. Substantial completion will be awarded only when the system is providing usable service to the Owner. If the system is commissioned in phases, the Contractor may request substantial completion for the completed phases. This definition is used to determine all warranty dates.

5.05 Warranty/Support Program

The control system manufacturer shall supply a one (1) year parts and labor warranty and comprehensive support program for all items and software supplied under this section (except as noted below). Power surges and lightning damage shall be regarded as an act of nature and not included as part of the warranty. Acts of nature are not limited to these conditions only but also include the majority of natural events that result in operating conditions outside the limits of this specification. Acts of violence, vandalism, or other intentional damage are not covered by warranty or support.

The warranty shall begin from the time of "substantial completion" as determined by the engineer. The manufacturer shall provide a 24-hour response to calls from the Owner. The manufacturer, at his discretion, may dispatch replacement parts to the Owner by next-day delivery service for field replacement by the Owner or their designated representative. Any damage to the control system caused by the actions of the Owner in attempting these field replacements shall be the sole responsibility of the manufacturer. If, during the warranty period, satisfactory field repair cannot be attained by field replacement of parts by the Owner, the manufacturer shall dispatch "factory" personnel to the job site to complete repairs at no cost to the Owner.

All computer equipment including the server and OWS, keyboards, LCD display, printer, communications devices, and associated UPS shall be covered by a one (1) year warranty beginning with "substantial completion" or the standard manufacturer's warranty whichever is later. Consumables are not covered by warranties. Lightning damage shall be regarded as an act of nature and not be included as part of the warranty on these components.

5.06 Software Upgrades: The support period shall begin from the time of "substantial completion" as issued by the engineer. The support program shall include free updates of all SCADA application software and central server/RTU operating system software shall be provided for a period of 7 years. The updates may be installed by contractor personnel over the support link or via telephone conversation with customer personnel. After the warranty expiration, updates shall be free of charge to the customer, but an installation fee may be charged for legitimate labor costs and travel expenses if contractor personnel are required to install the updates.

5.07 Telephone Support: The contractor shall provide telephone support to the customer for as long as the owner has the system. An after-hours support phone number will be provided for non-business hours support.

5.08 Remote Interactive Support: The contractor shall provide remote interactive support to the customer as an customer initiated remote software service or customer provided software solution should they not prefer the contractor software for as long as the owner has the system. An after-hours support phone number will be provided for non-business hours support.